

Year 10
Pathways 1

Index Laws Mini Test

Name: Solutions

Total Marks: 20 marks
Total Time: 15 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Calculators Allowed
1/2 A4 page (one side) of handwritten notes

Question 1

2 marks

Evaluate, giving answers in simplest form without indices.

$$\begin{aligned} \text{(i)} \quad & (3^2)^3 \\ & 3^{2 \times 3} \\ & = 3^6 \\ & = 729 \end{aligned}$$

✓

$$\begin{aligned} \text{(ii)} \quad & 4^2 - 4^{-1} \\ & 16 - \frac{1}{4} \\ & = 15\frac{3}{4} \text{ or } 15.75 \end{aligned}$$

✓

Question 2

8 marks

Simplify each of the following expressing answers with positive indices where appropriate.

$$\begin{aligned} \text{(a)} \quad & \frac{(a^4b)^{-2}}{2(ab)^5} \\ & \frac{a^{-8}b^{-2}}{2a^5b^5} \checkmark \\ & = \frac{1}{2a^{13}b^7} \checkmark \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad & \left(\frac{9k^5}{n^7}\right)^2 \\ & = \frac{9^2k^{10}}{n^{14}} \\ & = \frac{81k^{10}}{n^{14}} \checkmark \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad & \frac{a^2b^0}{bc^0} \\ & = \frac{a^2 \times 1}{b \times 1} \\ & = \frac{a^2}{b} \checkmark \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad & \left(\frac{3a}{2b}\right)^{-2} \\ & \left(\frac{2b}{3a}\right)^2 \checkmark \\ & = \frac{4b^2}{9a^2} \checkmark \end{aligned}$$

$$\begin{aligned} \text{(e)} \quad & 4m^4n^{-1} \times 3n^{-5} \\ & 12m^4n^{-6} \checkmark \\ & = \frac{12m^4}{n^6} \checkmark \end{aligned}$$

Question 5

3 marks

Solve each of the following equations.

(a) $\frac{2.38 \times 10^7}{1.02 \times 10^3}$

$$= \frac{2.38 \times 10^4}{1.02}$$

$$= 0.05802 \times 10^4$$

$$= 5.802 \times 10^2 \checkmark$$

(b) $9^{2x} = 27^{x-1}$

$$(3^2)^{2x} = (3^3)^{x-1}$$

$$3^{4x} = 3^{3x-3} \checkmark$$

$$\begin{array}{r} 4x = 3x - 3 \\ -3x \quad -3x \\ \hline x = -3 \checkmark \end{array}$$

Question 5

6 marks

Evaluate the following

(a) $125^{\frac{2}{3}}$

$$(\sqrt[3]{125})^2 \checkmark$$

$$= 5^2$$

$$= 25 \checkmark$$

(b) $9^{\frac{1}{2}} + 16^{\frac{1}{4}}$

$$= \sqrt{9} + \sqrt[4]{16} \checkmark$$

$$= 3 + 2$$

$$= 5 \checkmark$$

(c) $(8x^9)^{\frac{1}{3}}$

$$= 8^{\frac{1}{3}} x^{\frac{9}{3}} \checkmark$$

$$= \sqrt[3]{8} \times x^3$$

$$= 2x^3 \checkmark$$